Capstone Progress Proposal #1

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The first two weeks of the semester have been used to plan out tasks for this capstone in a more concrete fashion, including rough estimates of timing. This plan is not exhaustive, but is meant to provide myself a framework for the coming weeks. Particularly the sections surrounding mathematical updates and experimentation. Additionally, starting work has been put in according to this plan. **Section 1.1 (Neo4j Port) has been completed, and section 1.2 (math / concept updates) has been started**.

1 System Implementation: Weeks 1 - 4

1.1 Neo4j Port

1. Move from a PyVis + Python approach to a Neo4j + Python approach
    - Allows for native graph query (far better than PyVis offers)
    - Allows for usage of relationship properties
    - Graph cycle friendly
    - Usable without a full ontology $\longrightarrow$ Contrary to initial suggestion in the proposal, a full ontology would likely add additional complexity which is unnecessary instead of leaving it as a property graph
2. Graph Model Setup
    - Properties: alpha, beta, reliability
    - Relationships: predicate, alpha, beta, confidence
    - Potentially store predicate priors as well
3. Does not support colors like PyVis did, but this is currently seen as an acceptable tradeoff, as it is far easier to query the entire system instead of relying on visual cues.

1.2 Mathematical Updates

1. Finalize the alpha/beta belief math
    - Explicitly define 'prior_strength' as a counter-weight with better bounds.

- Explore dampening strength based on node weight, as right now it is taken without dampening and without bounding

2. Address cycle amplification & feedback loop concerns

   - Create a reliability decay per loop
   - Cap contribution per update $\longrightarrow effective\_s = min(s, s\_max)$

3. Not a change, but a clarification for paper and to lock in the system design - explicitly clarify what is Bayesian and what is heuristic (alpha/beta updating, node weighting, predicate priors) and the rationale behind that. Will make final paper writing easier, as well as experimental design

2 Dataset Building: Weeks 5 - 6

1. Start with 1 dataset

   - Small & Expressive, potentially built off of Medical dataset from last semester
   - Include cycles, shared predicates, mixed confidence, etc.

2. Branch into 2 datasets

   - One with unique triples only (BKG)
   - One with repeat data / evidence (NARS)
   - Rationale: Don't attempt to force NARS onto the BKG. NARS needs repeat data to build confidence & frequency data. The second dataset may also be used in the BKG for a given experiment, but a dedicated NARS dataset is needed

3 NARS Implementation / Testing: Week 7

1. Choose a NARS Build - ONA or NARSPython likely

2. Test datasets specifically against NARS to make sure they're working properly, map subject, object, predicate, confidence properly against NARS variables

3. Understand how to access subject bag to get a full picture of all objects, relationships, and confidences in NARS

4 Experimentation: Week 8 - 11

1. Run experiments using the curated datasets to compare the BKG framework to the NARS system.

2. For the BKG, utilize logging of confidence trajectory, node reliability, and predicate priors (allows for plotting, trend analysis)

3. For NARS, utilize the concept bag for much of the same, to see how the entire dataset is being handled. For NARS, the confidence trajectory of concepts will be tracked as new evidence is incrementally applied.

4. Potential individual experiments

   - Ordering sensitivity: Enter data in different orders to see if it is dependent on order for confidence and reliability
   - Robustness to Contradiction: Enter directly contradictory evidence to see results
   - Graph Heavy Types: Cycle-Heavy data vs Tree-Heavy data, Sparse data vs Dense data

5. Remember that comparison between the frameworks and their results is about behavior, not architecture. NARS and BKG attempt to handle the same type of data, but do so in fundamentally different ways.

5 Writing, Tuning, Final Experiments: Week 12 +

This section is deliberately vague. Notes, sources, logs, experiment results, etc. will be taken across the entire project. Experiment re-runs may be needed, or additional parameters may want to be set or added. The timing here also allows for project delays.

Final report is expected to include comparative analysis between the two frameworks with evidence including logging, concept bags, and visualizations.